

AB
DEF → A

CK = AB
DEF → A

EK →

ABC → D
AB → D
C →
E →

CK = ABC

$\alpha \rightarrow \beta$

$\alpha \in \{SK, CK, PA, PAC, NPA, NPAC\}$

$\beta \in \{P, NP\}$

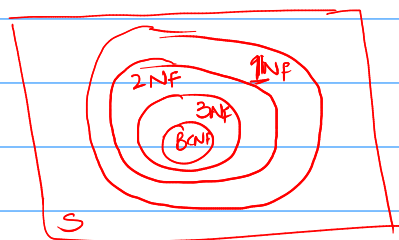
	1st NF	2nd NF	3NF	BCNF
SK → P	✓	✓	✓	✓
SK → NP	✓	✓	✓	✓
CK → P	✓	✓	✓	✓
CK → NP	✓	✓	✓	✓
P → P	✓	✓	✓	X
P → NP	✓	X	X	X
PC → P	✓	✓	✓	X
PC → NP	✓	X	X	X
NP → NP	✓	✓	X	X
NPC → P	✓	✓	✓	X
NPC → NP	✓	✓	X	X

System
must
have
FD

α is CK/SK
or
 α, β is prime
or
 $\alpha\beta$ is non prime

α is CK/SK
or
 β is prime

α must be CK or SK



$R(A B C D E F G H)$

$AB \rightarrow C$

$A \rightarrow DE$

$B \rightarrow F$

$F \rightarrow GH$

L	B	R
A	F	C
B		D
		E
		G
		H

$AB^+ \rightarrow ABCDEFGH$

$C.K = \{AB\}$

$PA = \{A, B\}$

$NPA = \{C, D, E, F, G, H\}$

	BCNF	3NF	2NF
$AB \rightarrow C$	✓	✓	✓
$A \rightarrow D$	X	X	X
$A \rightarrow E$			
$B \rightarrow F$			
$F \rightarrow G$			
$F \rightarrow H$			

INF

$R(A B C D E)$

$AB \rightarrow CD$

$D \rightarrow A$

$BC \rightarrow DE$

L	B	R
B	A	E
	C	
	D	

$B^+ \rightarrow B$

$BA^+ \rightarrow$

$BC^+ \rightarrow$

$BD^+ \rightarrow BDACE$

$PA = \{A, B, C, D\}$

$NPA = \{E\}$

$CK = \{BA, BC, BD\}$

	BCNF	3NF
$AB \rightarrow C$	✓	✓
$AB \rightarrow D$	✓	✓
$D \rightarrow A$	X	✓
$BC \rightarrow D$		✓
$BC \rightarrow E$		✓

3NF

R(A B C D E F)

AB → C

DC → A

DC → E

E → F

	BCNF	3NF	2NF
AB → C	X	✓	✓
DC → A		✓	✓
DC → E		X	X
E → F			

CK = {BDC, BDA}

PA = {A, B, C, D}

NPA = {E, F}

1NF

R(A B C D E)

CE → D

D → B

C → A

	BCNF	3NF	2NF
CE → D	✓	✓	✓
D → B	X	X	✓
C → A			X

L	B	R
C	D	A
E		B

CK = {CE}

PA = {C, E}

NPA = {A, B, D}

1-NF

R(A B C D E F G H)

ABC → DE

ABCD → EF

E → GH

H → G

G → H

CK = {ABC}

P = {A, B, C}

NP = {D, E, F, G, H}

	BCNF	3NF	2NF
ABC → D	✓	✓	✓
ABC → E	✓	✓	✓
ABCD → F	✓	✓	✓
E → G	X	X	✓
E → H			✓
H → G			✓
G → H			✓

2NF

R(A B C D E)

$BC \rightarrow A$

$BC \rightarrow D$

$BC \rightarrow E$

$D \rightarrow B$

BCNF

✓

✓

✓

x

3NF

✓

✓

✓

✓

3NF

$PA = \{B, C, D\}$

$NPA = \{A, E\}$

$CK = \{BC, CD\}$